The Economics of Wars and Their Consequences

Economic History · Lecture 6 · 16 October 2026 · Sources: lecture slides, lecture transcript, Juhász (2018), AER

Big picture

- WWI was financed by debt. Winners borrowed largely from the USA; the losers were charged reparations.
- Germany's total burden was about 299% of its 1913 GDP. Its tax revenue could not carry it.
- The central bank printed money; hyperinflation followed in 1923.
- Hyperinflations stopped abruptly once expectations changed credibly.

1. War economy and endowment

- Churchill: 1914–1945 is the "second thirty years' war"; the interwar period was a period without war, not peace (mistrust, no coordination).
- In WWI France and Germany spent more than 50% of GDP on the war. This was only possible for industrialised economies with a surplus. Today war is relatively cheaper ("bangs per buck", Ferguson 2001).
- Allies were larger in population, area and GDP. Central Powers could only win by surprise, not with resources (Blitzkrieg logic). Ritschl (2014): for Germany 1914 was the last chance to win.
- Rathenau's input-output planning model (WWI) was called by Lenin "a complete guide to socialism".
- Stronger suffering in WWI meant earlier women's suffrage (Germany, Austria: 1918–20; UK 1928; Switzerland 1971).

2. Reparations and war debt

Type	Purpose	Share of 1913 GDP
A	direct war damage	19%
B	Allied war debt to the USA (France, UK)	80%
C	punishment, home-front propaganda	152%
Total		**299%**

Keynes (1919): "Germany can never execute this treaty. Either renegotiation or anger and revenge!"

The London Schedule demanded more than 10% of German GDP per year. Dawes Plan (1924) and Young Plan (1930) lowered payments; Hitler stopped all payments in 1933.

3. Three ways to pay – and why printing money is the last resort

Option	Problem
Higher consumer taxes → current account surplus (IMF recommendation)	Consumers oppose it; workers' taxes would raise export prices
New debt (rescheduling)	Who lends to a country with 300% debt, and at what interest rate?

Option	Problem
Print money (seigniorage)	Devaluation, more expensive imports, falling real tax revenue, capital flight

Printing money is the last resort.

4. German hyperinflation 1921–1923

- 1920/21: reparations announced → political tension and falling tax revenue.
- Summer 1921: central bank starts financing the budget by printing money.
- 1922 budget (million Goldmark): revenues fell from 1,205 (Q1) to 646 (Q4); spending stayed at about 1,500 per quarter; deficit rose from 499 to 826.
- Quarterly price inflation: 55.8% (Q1), 308.3% (Q3), 419.9% (Q4) 1922.
- Jan 1923 onwards: France occupied the Ruhr (most industrialised region); the government financed a general strike by printing money → hyperinflation.
- November 1923: abrupt stop after currency reform (Rentenbank), Dawes Plan.
- Austria, Hungary and Poland had similar hyperinflations. Poland had no state tax capacity (three tax systems from three empires).

5. Why did hyperinflation stop overnight?

- Money supply kept growing after the stop (e.g. Austria: monetary base +20–25% per month during hyperinflation; growth continued afterwards). Hence money growth alone cannot explain the stop.
- Expectations: Lucas (1976) "Inflation is driven by expected inflation". Sargent (1982): the end of hyperinflation is evidence for rational expectations.
- Adaptive expectations: prices keep rising as in the past; disinflation is slow and costly. Rational expectations: a credible change makes stopping possible immediately (baker example: nobody raises prices if others don't).
- Credible regime change: League of Nations, independent central banks, binding limits on government borrowing, international loans, gold-exchange standard. Kydland & Prescott (1977): time-inconsistency problem → binding commitment is needed.

6. Fragile stabilisation and the Great Depression

- 1924 looked stable, but the Dawes Plan was only an interim solution, linked to unresolved war debts.
- Hot money flowed into Germany and Central Europe.
- Gold-exchange standard with wrong parities (pound overvalued, France undervalued), foreign currency as transmission channel, gold hoarding → imbalances amplified the Great Depression (next lecture).

7. WWI and women's labour supply

Gay (2023), France:

- Men per 100 women (aged 15–50) fell from about 98 before WWI to about 89 after it.
- Counties with more military deaths saw more married women working. Elasticity about 1 (roughly 0.6–1.05) in censuses 1962–2012.
- Effect transmitted from mothers to daughters, from mothers-in-law to daughters-in-law and via social interaction.

8. Unintended consequences: Napoleonic blockade

Juhász (2018):

- Continental Blockade (1806): trade with Britain was rerouted via Helgoland, Gothenburg, Gibraltar and Malta. Trade costs rose most in northern France (Seine-Maritime 86 → 1,282 land-km to London), least in the south (Pyrénées-Orientales 548 → 689 km).
- Question: did heavily protected regions build up cotton spinning? Method: difference-in-differences with spindles per 1,000 inhabitants.
- Result: 1803–1812 spindles rose from 380,000 to about 1.4 million; regions with higher trade costs expanded more (coefficient 33.47, significant). Placebo industries (wool, leather) show no effect.
- Long run: one extra spindle in 1812 → 2.1–2.7 more in 1840 and 4.7–4.9 more in 1887. Value added effect only in 1860. Author calls long-run results "suggestive".
- Mechanism likely learning-by-doing externalities (not tested). Infant industry argument works here because protection was unintended and the conditions (skills) were already in place.
- Lecture link: sanctions can protect domestic industries (Russian cheese example).

9. Key exam questions

1. Why could WWI be financed with over 50% of GDP, but not before 1914 or after 1945?
2. Why did the Central Powers lose the long war despite strong early positions?
3. What are the A-, B- and C-reparations and why was Germany's debt burden so high?
4. Why is printing money the last resort? Which costs does inflation bring?
5. Why did hyperinflation stop from one day to the next, even though money growth continued?
6. How does Juhász (2018) identify the effect of protection, and what are the main results?